FVS-ACD (AcadianGY) 14.2 refinement: release notes for PREREG16 to PREREG49 and the October 16 cross-version benchmark

Aaron Weiskittel, University of Maine. Draft of October 27, 2026 (revision g, built on revision f of October 26, 2026), for the `holoros/fvs-acd` repository.

Bottom line

From September 27 to October 5, 2026, the 14.2 refinement work ran the preregistered studies PREREG16 to PREREG45 (with PREREG28b) against the 14.1 benchmark record. **No engine file, coefficient or default of the shipped 14.1 was changed by any of them, and 14.1 remains the release.** The work produced an adopted scoring convention (volume under C1, or C2 where tags allow), an adopted scorer revision (r2, from PREREG37 on), one regional NS candidate that passes its benchmark gates but has not cleared a Bakuzis grid (v141LTfi), and named failures that locate the remaining NS and QC errors. The QC height anchor track is closed with the shipped 0.111 kept for 14.2: a field tag anchor (1.242) corrects QC height increment on the benchmark and passes a top height gate on measured heights against the 10% FIA measurement quality objective for tree length, but it fails the published TOPHT gate and does not clear the Bakuzis QC grid (PREREG38 to PREREG41), and the earlier QC height gates built on row index pairs are corrected by an erratum (PREREG42). The origin of 0.111 is now traced: no record fits it, and it was set by hand from the rounded console output of a July 2026 stand level top height diagnostic (PREREG43). A Bakuzis QC grid started from measured stand lists (PREREG44) does not clear 1.242 either, and its reported trajectories show that neither scalar gives a credible long run height path, which points 14.3 at the QC height increment form. PREREG45 tested that form for 14.3: a per year relative height multiplier, exp(a + b HT/ESI) fitted on fold A field tags, is the first QC arm to pass every benchmark gate on held out field tags, the published TOPHT gate included, but over 100 years its top height still rises past the measured stand frontier and it adds two Reineke flags, so it is not a 14.3 candidate as specified. An independent red team review and four stress tests confirm that reading under a corrected top height offset, at both bootstrap limits of b and with ingrowth held at 1. The NB counterpart, PREREG45b, could not be decided on stand level measured top height and is not adopted. Each gated verdict follows a decision rule fixed before any held out value was read, and the one post hoc step, the PREREG21 split of its survivor term, is declared as such in that RESULT.

Arms ran as rescoring, offline recomputation, or copied runners whose new switches default to 14.1 behavior. The L and T components were not re-estimated, GT1 stays a named failure, and a "candidate" is a regional usage setting beside the NB and QC anchors, not part of any release.

A cross-version benchmark on October 16, 2026 compared 13.1h, 14.0.4 L2 and 14.1 on the 19 frozen benchmark sets, on a new set of 827 post-harvest Maine FIA stands, and over 200 yr no-harvest projections of 918 unmanaged Maine FIA stands. It changed no engine file either. Version 14.1 is the best ten year model (median absolute biomass bias 3.8% across the 19 sets, against 9.6% for 13.1h), yet it overstates biomass recovery after heavy removals by 1.39 Mg ha^{-1} yr^{-1}, double the observed rate, and it projects a plateau and decline in 64 to 84% of stands in each of the six named forest types, where the FIA curves show a peak only in spruce-fir. These

results point 14.3 at three changes, namely a thinning response limited to partial removals, senescence conditioned on forest type, and a repair of the diameter growth shortfall common to every version. The frozen 14.2 candidate arms p38_fit and qc_rh, run on its three Maine frames on October 26, 2026, reproduce 14.1 exactly, as both act only on Quebec heights.

On October 26, 2026 two more QC studies ran, both for 14.3 only. PREREG46 tested a QC height increment multiplier with a zero in relative height and found no candidate: the form with its zero fixed at the measured maximum could not be fitted with a declining multiplier, and the linear form fitted its zero at 1.64 times ESI, past every measured stand, its held out increment ratio near the frontier is 1.341 with an interval too wide to bound, and, like PREREG45, lets top height rise past the frontier over 100 years. The PREREG47 reading of absolute Bakuzis gates on the measured QC stand lists finds every arm read not credible over 100 years: 14.1 holds QC top height nearly flat after its first decades, and p38_fit, PREREG45 and the PREREG46 linear form all overshoot the measured frontier. 14.2 still keeps 0.111.

Two more studies for 14.3 ran on October 26 and 27, 2026. PREREG48 split the QC stems deficit of about 24 points on qc_c fold B into survivors, upgrowth, recruitment and residual on field tags. The deficit is ingrowth past the scoring floor, on both 14.1 (ingrowth share 1.269) and qc_rh (1.251), while the model keeps about 9% more survivors than the tags show (survivor ratio 1.094 and 1.089, a borderline pass of the 0.90 to 1.10 window), so rule 3 names a 14.3 ingrowth track, and the arm difference of 0.32 points shows it does not depend on the height form. An independent recomputation reproduced every point value (the bootstrap intervals were not recomputed) and found the cause: engine ingrowth enters near 5 cm and only 7 of 6,042 ingrowth records reach QC scoring floors of 9.1 to 33.7 cm within the interval, although the engine adds about as many stems in total as the field tags gain, so the track starts with ingrowth entry size and floor handling rather than the ingrowth rate. PREREG49 fitted a gamma frailty survival step for NS on fold A (theta 3.558, bootstrap interval -0.642 to 9.843). It brings the long interval survivor ratios into the window (0.930 on ns_c and 0.993 on ns_long_c, against 0.868 and 0.921 under v141LTfi) but predicts annual survival too low in 5 of 11 interval classes and fails the PREREG22 gates E3 and E6b and the C1 accuracy margin, so it is not adopted (rule 2). On the NS grid from measured stand lists, v141LTfi is not cleared against v141LT: it adds Eichhorn and top height monotonicity flags, and its 100 year top height is a median 15.6% below v141LT (26 of 36 stands lose 14% or more), inside the PREREG26 range, so the loss is not a synthetic start artifact. Neither study changes 14.2. 14.3 PREREG2, which re-estimated the form of the thinning term T, ran on October 27: the selected form C2 passed every harness gate but failed the survivor gate on mefia_ph_c, so it is not adopted and 14.3 keeps the 14.1 T.

Beside the benchmarks, 13.1h and 14.0.4 L2 were run on the 93 Howland Research Forest plots with the runner, inputs and ESI 12, 15 and 18 m of the 14.1 run of October 23. At mid site over 50 years, biomass changes by +29.7% under 12.3.6, +8.3% under 13.1h after a drop of 13.8% by year 10, -4.2% under 13.1 ship18, -11.0% under 14.0.4 L2 and -8.3% under 14.1, and the 14.x engines lose basal area from 41.5 to about 31 m^2 ha^{-1}, with 55 to 67 of 93 plots losing biomass. Only 12.3.6 matches the sustained gain that repeat ALS reports at Howland. 12.3.6 and 13.1 ship18 report Jenkins biomass on climate site index and the other three the runner's biomass on ESI, so only the changes compare (`results/howland_20261026/`).

Volume scoring convention and scorer r2

Observed benchmark volume used measured height where it exists, while predicted volume used model heights, so PREREG16 rescored every set under C0 (as published), C1 (the generic `himp` height curve on both sides) and C2 (measured height on both sides for tag matched survivors). Under 14.1 the C0 over C1 gap exceeded 3% in 12 of 20 sets, led by baxter_c (+13.9%), the frozen MNAP TNC set (+10.4%) and MNAP State (+9.6%), so the preregistered rule fired. The gap partly carries model height error, since where every height is measured mefia_l_c shows none (0.0%) but MNAP State still shows +9.6%. All 14.2 volume gates therefore use C1, or C2 where it exists, and height is judged separately through top height (TOPHT) and tag matched height increment. Restating the 14.1 record under C1, as PREREG16 required, changes five readings most (Table 1, full table in the PREREG16 RESULT).

Table 1. 14.1 (v141LT) total volume bias, percent, under C0 as published and under C1 (PREREG16).

Set	C0	C1
baxter_c	+27	+12
MNAP State	+0.6	-8.2
ns_long_fl	+1.7	-4.5
pef_fl	-14	-19
silc_l_c (wide interval)	-4.7	+3.6

PREREG35 built scorer r2 by insertion only (48 lines added, 0 removed), adding a defect gate, an exclusion check and an “established at t1” stratum row. Its identity gate failed on QC C2 alone, so no adoption verdict was drawn, and PREREG36 traced that failure to the frozen PREREG16 QC C2. The QC tag frame used there was a copy of the run frame, where TREE is a row index, so C2 had paired t1 and t2 records by row position, and only 38.6% (qc_c) and 15.6% (qc_fl) of those pairs are the same tree under the MAGPlot field tags. Correcting to field tags moves QC C2 by 0.003 to 0.006 (screened qc_c 1.073 to 1.079), and no gate from PREREG17 to PREREG35 read QC C2, so no verdict changes. With every PREREG36 gate passing, r2 scores 14.2 gates from PREREG37 on, and the field tag values supersede the PREREG16 QC C2 entries.

NS height form, ingrowth multiplier and IMULT

PREREG17 tested the NS ingrowth multiplier (IMULT) under threshold matching, and its stems criterion chose m* = 2.5, where E3 and E6b failed. PREREG18 refined the grid between 1.5 and 2.25 on a held out C1 volume criterion and selected 1.5, the lower edge, where every gate other than Edge passed (ns_c fold B volume ratio 1.0074, interval 0.9969 to 1.0179). An edge value cannot be adopted by design, and since the consecutive volume criterion asks for no more ingrowth than 1.5 while stems balance near 2.5, no flat NS scalar satisfies both frame types. Maine stays at 1.0.

PREREG20 fit a flatter two segment height increment correction for NS trees 25 cm and over (b1 0.034416, b2 0.005072). Every height gate passed (ns_c fold B large tree increment ratio 1.250 under 14.1 to 0.792, ns_long_fl TOPHT bias +5.47% to -1.75%). However, E6b failed by 0.08 points on ns_fl, so the form was not adopted alone. PREREG22 paired the form with IMULT 1.5

and every gate passed, which makes v141LTfi the first NS arm in the 14.2 series to pass H1, H2, S1, E3 and E6b together. Stems still run 4 to 12% low on the held out and long frames. PREREG25 showed that the remaining small tree deficit is a plot program effect, as v141LTf under predicts NS_RS height increment by about 40% at every size while NS_OP stands show no small tree deficit. No small tree size multiplier enters 14.2.

NS_RS and NS_OP programs and long interval checks

PREREG27 found that the long interval stem deficit is survivor loss, confirmed on held out sites. The fold B survivor ratio at 16 to 60 years was 0.868 on ns_c and 0.921 on ns_long_c. Predicted annual survival sits at 0.987 to 0.991 throughout, whereas observed survival rises from 0.980 to 0.983 at 5 years to 0.992 to 1.000 beyond 20 years, so flat IMULT 1.5 stays and NS survivor mortality over multi year steps is named for a later track.

PREREG29 refit the PREREG20 form on NS_OP alone, and the refit failed A1 and A2, so the pooled form stays. No production stand state removes the NS_RS deficit on held out sites, since the best quartile of any variable reaches only 0.82. The programs differ instead in plot size (100 to 455 m^2 against a fixed 400 m^2), diameter floor and height subsampling, and NS_RS is named as outside the 14.2 NS height calibration.

PREREG30 checked end of interval height on the long NS records. Every gate passed on ns_long_c fold B (pooled level ratio 1.0152 under 14.1 to 1.0041 under v141LTf, 1.1194 to 1.0734 at 20 years and over, TOPHT bias +2.37% to +0.41%). The longest intervals are 35 years (ns_long_c) and 40 years (ns_long_fl), so the PREREG26 top height loss is read as a synthetic start artifact, although its size at 100 years is not tested. The one exception, the ns_long_fl 10 to 19 year bin (0.9353 for 14.1, 0.9186 for v141LTf), was resolved by PREREG34 as a program effect. That bin is 62% NS_RS trees, program composition carries a share of 0.556 of the gap, and on its NS_OP stands the form moves the level from 1.0197 to 0.9964, as it does on ns_long_c.

QC height anchor family

PREREG19 scored QC_AFAC from 0.030 to 1.000 under C1. The screened qc_c fold A C1 ratio spanned only 0.0067 across the grid, below the 0.010 threshold, so the anchor is a height calibration and the QC volume excess sits in diameters and stems (stems about 24% low on fold B at every anchor). These notes use revision 3 of that RESULT, which keeps the revision 2 text, withdraws its C2 numbers and corrects the attribution of its tag match counts (PREREG42). PREREG23 refit the anchor on tag height increment. On its tagged survivors the shipped 0.111 predicted about a tenth of observed increment (fold B pooled ratio 0.086), and the candidate 1.061 lifted the ratio to 0.828 but moved the 25 cm and over class from -0.021 to +0.099 m yr^{-1}, worse by 0.077. PREREG37 later showed that these survivors, like those of PREREG28 and PREREG28b, were paired on row index frames, and under field tags the PREREG42 erratum reverses this verdict: the 25 cm and over class is better than 0.111 by 0.108 m yr^{-1} and every gate as PREREG23 coded it passes. Under the stricter PREREG38 window (pooled ratio 0.90 to 1.10) 1.061 sits just outside at 0.898, so it was not reopened. PREREG24 was a QC stems and QMD diagnostic, and its RESULT is not on the host, so neither its verdict nor its numbers is cited.

PREREG28 split the anchor by size (1.265 below 25 cm, 0.030 at and above) and PREREG28b kept 0.111 for large trees. Both passed every gate on screened qc_c fold B (pooled ratios 0.875

and 0.883), and under the preregistered choice rule the TOPHT difference (+11.64% against +11.97%, 0.33 points) fell under the 0.5 point threshold, so PREREG28b was named the QC candidate, at a TOPHT cost from +6.29% to +11.97%. PREREG33 reassigned the anchor from current DBH each year and kept every benchmark gate (TOPHT +11.14%). Under field tags the PREREG42 erratum leaves all three verdicts unchanged, but their fitted values rest on row index pairs.

PREREG37 asked what in the 14.1 height increment produces that size pattern, and found that the pattern is not a property of the increment. The PREREG23 helper, reused unchanged by PREREG28 and PREREG28b, read the run frames where TREE is a row index, the defect PREREG36 found for QC C2. Under MAGPlot field tags (737 fold B records against 301 by row index), observed over unanchored predicted increment is 1.169 pooled and 0.943 (0.643 to 1.235) at 25 cm and over, against 0.271 by row index, and the size contrast S is 1.265 against a preregistered threshold of 3.376. On field tags a single QC_AFAC near 1.17 fits, not 0.111. A weaker real size structure remains under field tags, which the H1 rule attributes to the height term, and the unanchored equation still reaches only 0.62 to 0.81 of ESI at age 50 from a seedling start. PREREG37 changed no frozen result. These notes no longer treat p28b_small as the 14.2 QC candidate, because its fit rests on row index pairs (PREREG37). The PREREG33 reassignment, which was never cleared, rests on the same pairs. The same evidence bears on the shipped 0.111.

PREREG38 refit a single anchor on fold A field tags (1.242). On fold B it brings the pooled predicted over observed height increment ratio to 1.053 (0.093 at 0.111), every diameter class within 0.04 m yr^{-1} of zero and the dominant cohort to 0.994 of measured t2 height (0.898 at 0.111), but the published TOPHT bias is +16.21% against a limit of +14.63%, so it was not adopted. PREREG39 traced the conflict to the observed height rule: 60.6% of observed TOPHT heights come from the generic `himp` curve, which gives 0.892 of measured height, and that step carries 9.29 of the 16.78 point gap between TOPHT and the dominant cohort. PREREG40 rebuilt the top height gate on measured heights against a 10% ceiling, the FIA field guide measurement quality objective for tree length (confirmed October 5, 2026): the 1.242 arm is +1.60% (-0.05 to 3.25) and 0.111 is -8.45%, and PREREG41 found the reading robust to tree selection on the same trees on both sides (+1.50% and -8.82%). The ceiling is loose, since every arm tested passes it.

The QC anchor track is closed with 0.111 kept for 14.2 (`2026-10-02_v142_qc-anchor_CLOSEOUT.md`). On measured tagged trees 0.111 predicts about a tenth of observed height increment, the dominant cohort reaches 0.90 of measured t2 height and measured top height is 8 to 9% low, so QC height and height driven outputs from 14.2 are biased low over long projections. PREREG42 also issued revision 3 of the PREREG19 result, which corrects the attribution of its tag match counts: 11,632 (1,569) are row index counts on the 378 qc_c stands that ran, and field tags give 10,226 (2,299).

PREREG43 and its addendum 2 traced where 0.111 came from. Two independent searches of 42,671 files on the analysis host, and a further search of the reachable Cardinal folders (8,905 files) and the `holoros/fvs-acd` history up to July 10, 2026 (63 commits), found no record that fits 0.111. The repository VERSION_HISTORY places it in the AcadianGY 12.9.2 operations recipe. A July 2 diagnostic on v12.7.0 output (`topht_recal_diag.R`, commit a259273d) printed the Quebec ratio of mean predicted to mean observed stand top height increment as 8.89, and a July 4 v12.10.1 stress test note (commit 4af1b2da) set the Quebec increment scale to the observed over predicted ratio, 0.111, without a computation. Addendum 3 read the

diagnostic's output file: the unrounded ratio is 8.888861 (2,472 stands), whose inverse is 0.1125, and the New Brunswick inverse is 0.5772, not 0.579, so neither shipped value is the inverse of its ratio. Both equal the ratio of the two means as printed to three decimals (0.018 / 0.162 = 0.111 and 0.135 / 0.233 = 0.579), so they were most likely computed by hand from rounded console output; this reading was not preregistered. 0.111 is therefore an operations setting from a stand level top height correction, carried unchanged through the v12.10.1 freeze into 14.1, and not a value fitted on QC tag pairs. The 14.2 decision does not change.

PREREG45, for 14.3 only, replaced the QC scalar with m(r) = exp(a + b r), r = HT/ESI, reassigned to every tree each year. Fitted on the 698 fold A field tag survivors, a = 0.9595 and b = -1.1239, so m falls from 1.97 at r 0.25 to 0.74 at r 1.125. On screened qc_c fold B (737 field tags) the pooled predicted over observed increment ratio is 1.003 (0.897 to 1.143), the three diameter class biases are +0.002, +0.002 and -0.014 m yr^{-1}, every relative height bin is within 0.04 m yr^{-1} (worst +0.030), the published TOPHT bias is +14.14% against the +14.63% limit and the dominant cohort reaches 0.980 of measured t2 height, so every benchmark gate passes, the first QC arm to do so. From the 36 measured PREREG44 stand lists, median top height over ESI goes from 0.784 to 1.163 at 50 years and 1.309 at 100, above the 99th percentile of measured QC stands (1.175) in 32 of 36 stands and still gaining about 1 m per 20 years, and Reineke flags two stands that 0.111 does not. The verdict is Rule 4, not cleared on long run grounds. The multiplier is positive at every relative height, so it slows height growth without stopping it; a form with a zero or an asymptote is the evident next arm. qc_rh also removes most of the top height falls that flag 0.111 (9 stands against 36). Revision 2 of the PREREG45 result adds an independent red team review, which re-derived every number, and four post hoc stress tests. A trace of every multiplier call matches exp(a + b HT/ESI) to 7.8e-15 (the registered a = b = 0 identity test could not show this). With the 1.9 m first year engine height jump removed, the median is 1.205 and 24 of 36 stands exceed the frontier. At the bootstrap limits of b the medians are 1.239 and 1.378, and with ingrowth held at m = 1 it is 1.309, so gate C fails in every variant. The benchmark pass is lenient above HT/ESI 1, where a bias of +0.030 m yr^{-1} is a ratio near 1.4 and fold A holds only 58 trees, and the two Reineke flags come from faster growth on unchanged mortality rather than from the height form.

PREREG46, for 14.3 only, tested two forms with a zero in relative height, fitted with the increment integrated over each tag's interval: qc_lz, max(0, a + b r), and qc_ez, exp(a + b r) times a taper to zero at r 1.285, the largest measured top height over ESI. qc_ez stopped at its fit: with the zero fixed at 1.285 the fold A estimating equations give b positive (a 0.4303, b 0.9219), a registered stop, so it was not run. qc_lz was run under three logged deviations that drop qc_ez from the chain. It fits a = 2.2292 and b = -1.3631 (bootstrap interval -2.062 to -0.729), so its zero is at r 1.635 and m is still 0.48 at the largest measured relative height. Every harness gate passes, the fold B pooled ratio is 1.047 and the published TOPHT bias +14.45% against +14.63%, but the predicted over observed ratio in the 1 to 1.25 relative height bin is 1.341, with a bootstrap upper limit of 2.680 against 1.50, so the two relative height bin gates registered for this study fail (Rule 2, not adopted). From the 36 measured stand lists its median top height over ESI reaches 1.285 at 100 years (1.182 with the ST2 engine offset removed, 19 of 36 stands over 1.175), and it adds Reineke flags, so no candidate is named.

PREREG45b took the NB arm to stand level, because NB tree numbers are not stable tags (PREREG32): observed measured top height increment against the cohort increment predicted at NB_AFAC 1. On 70 fold A stands b is -2.19 with a bootstrap interval from -11.0 to 1.06. On fold B

the pooled ratio is 0.556 (0.403 to 0.835), outside 0.90 to 1.10, so Rule 2 applies: not adopted. 14.1 itself predicts only 0.281 of the fold B measured increment while its published TOPHT bias is +1.04%, and observed increment differs by a factor of 1.7 between folds, so the measured top height of NB frames, which changes with the subset of trees measured, is too noisy a target. NB stays at 0.579, and a tree paired NB height sample is the prerequisite for any 14.3 NB height arm.

MNAP benchmark and State height

PREREG21 explained why MNAP State model heights run tall and placed the whole gap on the model side (H_p 1.089 against H_o 0.993). At t1, 32% of trees carry no measured height and finish 21% tall at t2, while trees starting from a measured height finish 5.5% tall, because the model grows them about 2.4 times faster than the remeasurement shows. The gap splits roughly 55 to 45 between t1 imputation and survivor increment, and the study was diagnostic.

NB height source and the empty t2 audit

PREREG32 audited the 38 scored stands admitted with an empty observed t2 list, 33 of 355 of them in nb_c. All 38 are plots remeasured with live stems, every one below the 5 cm scoring floor, so none is harvested, lost or a frame defect. Excluding them is material for nb_c only, where C1 volume barely moves (0.956 to 0.954) but stems fall from 1.187 to 1.140, as predicted ingrowth past the floor on these plots is a real over prediction. The audit kept the published admission and proposed three scorer checks, which r2 now carries, and the nb_c reading of the earlier NB height source study (PREREG14 steps 1 and 2) would move only from 0.987 to 0.985.

Bakuzis grids

PREREG26 ran the Bakuzis NS grid for v141LTfi against 14.1 under a noninferiority rule and did not clear it. The worst basal area site ordering margin is 8 inversions against 7, one inversion in one cell of pure balsam fir (*Abies balsamea* (L.) Mill.) at 6,000 trees ha^{-1}, which the follow up arms attribute to threshold matching with IMULT 1.5. The reported arms add a larger caution, as the height form lowers 100 year top height by 14 to 25% (balsam fir on an 18 m ESI, 16.2 m against 20.8 m under 14.1).

PREREG31 ran the Bakuzis QC grid for p28b_small and did not clear it, with four new site ordering flags and worse margins on Reineke (1.496 against 1.266) and basal area ordering (14 inversions against 13). The reference is not a credible long run projection, since under 14.1 every seeded tree keeps 0.111 for 100 years and top height at age 50 is 0.26 to 0.80 of ESI. PREREG33 failed the same grid (Reineke 1.412 against 1.266). Both grids tested anchors fitted on row index pairs (PREREG37). PREREG41 ran the same grid for the field tag anchor 1.242 against the frozen 14.1 trajectories and did not clear it either: three new basal area site ordering flags and a Reineke margin of 1.490 against 1.266, the same pattern as p28b_small. The margins are small and appear tied to the seedling start (stems rise from 1,000 to about 2,800 ha^{-1} through ingrowth before any self thinning) and to the 2.5 m span of the QC ESI levels, but the rule was fixed before the grid was run and is not waived. This verdict does not depend on the PREREG40 limit.

PREREG44 repeated the comparison from measured qc_c t1 stand lists (36 fold B stands stratified by ESI and stand density) instead of a seedling cohort, with Reineke evaluated only after a stand reaches half the QC frontier and site ordering tested on ESI bin means, and with the decision fixed in advance to inform 14.3 only. Every harness gate passed, including identity gates showing the driver edit left the earlier modes unchanged. The field tag anchor 1.242 did not clear: three new Reineke flags (worst margin 1.101 against 0.768), two new yield density flags and a larger worst top height fall (4.87 m against 1.04 m). The reported trajectories show why a scalar does not settle the question. From a mean measured top height of 14.49 m, 0.111 reaches only 17.29 m after 100 years (0.92 of ESI) and loses top height at some point in every stand, while 1.242 reaches 28.01 m (1.50 of ESI) and is still rising.

PREREG47 set absolute gates for the same grid, registered before any threshold or PREREG46 long run output existed: gate C on the measured frontier, a credibility floor, early and late top height growth (MG, MGN, LG), top height monotonicity with thresholds from measured stands, strict site ordering and the PREREG44 relations. Its thresholds were computed once on October 26, 2026 (F_HT 9.52 m, S_HT 11 stands); the early growth floor MG_LO came out at -0.025 m yr^{-1}, so the early growth gates can fail only an arm whose top height shrinks. Every arm read fails Rule 2. 14.1 fails late growth, with 7 of 36 stands gaining in every 20 year window after year 21 and late window medians within 5 cm of zero. p38_fit (1.418), PREREG45 (1.205) and PREREG46 qc_lz (1.182, read blind) all fail gate C on the offset removed median against 1.175. Strict site ordering fails in every arm, qc141 in 3 of 6 cells and the others in 6 of 6.

Cross-version benchmark, October 16

The benchmark asked how three versions score over a ten year hindcast, how well each recovers biomass after a recorded harvest, and how each behaves when unmanaged stands are grown for 200 years without harvest, since FVS-ACD now carries 100 year projections in the FCCL-II and NCASI multi-model efforts. The versions were 13.1h (AcadianGY 13.1 with hierarchical ingrowth and mortality posteriors, the Seven Islands configuration), 14.0.4 L2 (the 14.1 overlay sourced with both gates off) and 14.1 (L and T on), all run through the frozen benchmark runner on OSC Cardinal with md5 checked engines. On Maine plots 14.1 equals the 14.2 reference arm qc141, because the Quebec height anchor acts only on Quebec stands. The frozen candidate arms p38_fit and qc_rh change only the Quebec height path and should equal qc141 on Maine plots as well, and on October 26, 2026 they were run on the same three frames (OSC Cardinal, jobs 15885111 to 15885113, the engine code copied from firebreather): every compared value of results, tree level predictions and trajectories equals 14.1 exactly. Biomass is NSVB aboveground dry biomass of trees at least 5 cm dbh, bias carries a 95% stand bootstrap interval from 1,000 resamples, and the equivalence regions (intercept within 25% of the observed mean, slope within 1 plus or minus 0.25) were fixed before any new output was read. Rescoring the frozen 14.x runs reproduces the published per set biases to within 0.0005 percentage points on the 72 of 114 rows where the paired stand sets match.

Across the 19 frozen sets (28,898 paired stands), 14.1 has the smallest median absolute bias for total volume (3.7%) and biomass (3.8%), against 3.9% and 5.3% for 14.0.4 L2 and 10.3% and 9.6% for 13.1h, and it passes biomass equivalence in 12 sets against 13 for 14.0.4 L2 and 10 for 13.1h. On the 918 unmanaged Maine FIA stands every version underpredicts biomass, from -9.6% (-11.5 to -7.8) for 13.1h to -11.7% (-13.5 to -10.0) for 14.1 and -13.7% (-15.5 to -12.0) for 14.0.4

L2. Top height there is nearly unbiased (+0.4 to +0.9%) while quadratic mean diameter is low by 6.3 to 12.0%, which places the shortfall in diameter growth and in survival of larger stems.

The 827 post-harvest stands, classified with the FCCL common practice rules, separate the versions most clearly (Table 2).

Table 2. Predicted minus observed annual aboveground biomass change after harvest, Mg ha^{-1} yr^{-1}, with 95% stand bootstrap intervals (cross-version benchmark, `acdx_postharvest_by_type_DATA.csv`).

Harvest type	Stands (n)	Observed (Mg ha^{-1} yr^{-1})	13.1h	14.0.4 L2	14.1
Light	146	2.53	-1.11 (-1.68 to -0.59)	-1.44 (-1.89 to -0.95)	+0.46 (-0.13 to 1.00)
Intermediate	191	2.06	+0.37 (0.07 to 0.68)	-0.40 (-0.71 to -0.08)	+1.41 (1.10 to 1.71)
Heavy	221	1.39	+0.69 (0.46 to 0.95)	+0.23 (-0.03 to 0.48)	+1.39 (1.13 to 1.64)
Low volume	269	2.85	-0.14 (-0.39 to 0.11)	-0.64 (-0.89 to -0.38)	+0.73 (0.47 to 1.01)

Since 14.0.4 L2 differs from 14.1 only by the L and T terms and comes closest after heavy removals, the excess under 14.1 implicates the thinning response T. It was fitted on commercial thinning (CTRN and BAXTER) and appears to treat an overstory removal as a thinning, so a residual stand of saplings and poles receives a release meant for a thinned pole or sawtimber stand.

Over 200 years without harvest the two families fail in opposite directions. Version 13.1h declines in essentially no stand of any forest type (0 to 2%), and in spruce-fir, the one type whose FIA curve peaks (near 104 years), it ends at 2.33 times the curve. Under 14.1, 64% (spruce-fir) to 84% (elm-ash-cottonwood) of stands in the six named forest types plateau and decline, with median peaks at ages 122 to 139, so 14.1 reproduces the spruce-fir decline but imposes one on hardwood and pine types where the FIA record shows none. By year 200, 14.1 sits at 0.56 of the maple-beech-birch curve and 0.70 of the white-red pine-hemlock curve, and for stands older than 120 years it projects net change of -0.03 and -0.11 Mg ha^{-1} yr^{-1} in spruce-fir and maple-beech-birch against an observed 1.41 and 1.48.

For 14.3 this evidence supports three changes, none of which is preregistered yet. First, the thinning response should be limited to partial removals, for example by residual relative density or removal fraction, and the post-harvest set is now a ready benchmark for that. Second, the senescence that produces the decline in 14.x should be conditioned on forest type, since only spruce-fir declines in the FIA record. Third, the diameter growth shortfall on unmanaged FIA stands, about 10% in quadratic mean diameter, is common to all three versions and is the largest single source of ten year bias in Maine. The full report, tables, figures and scripts are in `docs/benchmarks/2026-10-16_cross-version/`.

Current 14.2 candidates

- **Scoring.** Volume gates under C1, or C2 with field tags, height scored separately, scorer r2 from PREREG37 on.
- **NS.** v141LTfi, the PREREG20 height form with threshold matching at IMULT 1.5. It passes the benchmark (PREREG22), and its height form (v141LTf) passes the long interval height checks (PREREG30, where v141LTfi is reported and matches it within 0.0006, and PREREG34). It does not pass the PREREG26 Bakuzis grid, it is not cleared on the NS grid from measured stand lists either (PREREG49), and NS_RS is outside its calibration.
- **QC.** None. 14.2 keeps the shipped QC_AFAC 0.111; the field tag anchor 1.242 passes the PREREG38 height increment gates and the PREREG40 measured top height gate, but fails the PREREG38 published TOPHT gate and does not clear the seedling start Bakuzis QC grid (PREREG41), and the track is closed (PREREG42 close out note). A Bakuzis grid from measured stand lists does not clear 1.242 for 14.3 either (PREREG44). For 14.3, the relative height multiplier of PREREG45 passes every benchmark gate but is not cleared on long run grounds; its NB counterpart (PREREG45b) is not adopted. The zero form of PREREG46 names no candidate, and the PREREG47 reading finds no QC arm read credible on the long run.

By its own preregistration, v141LTfi does not enter a release before a Bakuzis grid clears it.

Open items

- The QC height increment form for 14.3, since from measured stand lists 0.111 stalls top height and 1.242 lets it rise to 1.5 times ESI and still climbing at 100 years (PREREG44). PREREG37 already shows why: on field tags observed over predicted increment falls from 1.655 below HT/ESI 0.5 to 0.523 at 1 to 1.25, while the coded `dht` has no relative height term. PREREG45 fitted a relative height multiplier that corrects the increment on held out tags but, being positive at every relative height, does not bound top height; the next arm was a form with a zero or an asymptote. Stress tests confirm the reading (PREREG45 revision 2). PREREG46 tested that arm and named no candidate, and PREREG47 finds a flat 14.1 and three alternatives that overshoot (a bracket that is an inference to test, since no arm passes), so the next QC study needs height increment data near the frontier or a shape constraint chosen from outside the tag set. The NB arm at stand level was not adopted (PREREG45b); NB needs a tree paired height sample first.
- Whether noninferiority to a 14.1 reference that cuts seedling cohort height growth to about a ninth is the right Bakuzis gate for QC.
- NS survivor mortality over multi year steps, or a long interval remeasurement convention (PREREG27). The PREREG49 frailty step fixes the long interval survivor ratios but not annual survival by interval, so a form keyed on stand state, or the scorer convention, remains to test.
- The 100 year size of the NS height form top height loss, which PREREG30 could test only to 35 years. PREREG49 measured it from measured stand lists: a median 15.6% at 100 years and 8.9% at 35 years, so the loss is not a seedling start artifact; part of it is v141LT running above the NS measured frontier (4 of 36 stands at 100 years, none under v141LTfi), and whether the rest is a v141LTfi height form deficit is not settled.

- The 7.0 m old harness ESI offset between ns_c and ns_long_c (PREREG29), reported for repair and not acted on.
- MNAP State t1 height imputation, large tree increment and the unread mefia_l_c model side control (PREREG21).
- The QC stems deficit of about 24 points shared by every anchor arm (PREREG28). PREREG48 (October 26, 2026) places it in ingrowth past the scoring floor, traced to engine ingrowth entering near 5 cm below QC floors of 9.1 to 33.7 cm; the 14.3 track is ingrowth entry size and floor handling, and whether the QC scoring floor itself should be revisited.
- The PREREG24 RESULT and the PREREG23 and PREREG24 preregistration texts, which are not on the analysis host.
- A 14.3 thinning response limited to partial removals, since 14.1 overstates biomass recovery after heavy removals by 1.39 Mg ha^{-1} yr^{-1} and after intermediate removals by 1.41 (cross-version benchmark). 14.3 pre-registration 1 (October 18 to 19) adopted no eligibility rule; 14.3 pre-registration 2, which re-estimates the form of T, was registered on October 26, 2026 and ran on October 27, 2026: the selected form C2 is not adopted (rule 3; survivor ratio 0.867 above d/Dq 1.5 on mefia_ph_c), so the T form remains open, and the sealed NH and VT confirmation is open (16 heavy stands, 68 of 108 on climate site index).
- Senescence conditioned on forest type for 14.3, since 14.1 declines in 64 to 84% of stands in each of the six named types while the FIA curves peak only in spruce-fir (cross-version benchmark).
- The diameter growth shortfall on unmanaged Maine FIA stands, where quadratic mean diameter is low by 6.3 to 12.0% in every version (cross-version benchmark).

Reproducibility

Scripts, preregistrations, diffs and aggregate tables are on branch `v142-refinement-20260926` of `holoros/fvs-acd`, in commits 3e429c13 and 58f04eb2 (PREREG16), 793b0239 (PREREG17), 3aff1c88 (PREREG19 to 21), b05d58b5 (PREREG18, PREREG22 to 25 and the PREREG19 correction), 96697305 (the PREREG22 result), 5fe43221 (PREREG26 to 29), 6e42f330 (PREREG30 to 32), fe7898f7 (PREREG33 to 35), d93b4aad (PREREG36), 93ac8efe (PREREG37 and the 1 October notes), 01998dd8 (PREREG38 and PREREG39), 09c6bfa3 (PREREG40), 2693d895 (PREREG41), a03a7702 (PREREG42, the erratum, the PREREG19 revision 3 and the QC close out note), c818b99c (the 2 October notes), 599c16a4 (PREREG43 and the PREREG44 draft), c4c2a572 (PREREG43 addendum 2, PREREG44 and the first 5 October notes) 7bbe6615 (PREREG43 addendum 3, the PREREG45 draft and the second 5 October notes) a666987c (PREREG45 and the third 5 October notes) fc78b2be (PREREG45 revision 2, the stress tests, PREREG45b and the fourth 5 October notes) c9ad8277 on branch `acdx-crossversion-20261016` (the cross-version benchmark package, revision e of the notes, the README status section and the VERSION_HISTORY Era 7 to 9 entries), and d06c68ed on branch `v142-p46p47-docs-20261026` (PREREG46, the PREREG47 reading, the 14.2 arms on the Maine frames and the 1.9.0 drivers, in `scripts/v14.2dev/2026-10-26/`); revision f is in 695b7fec on branch `v142-autopilot-20261026`, and revision g is in the commit after c2d79edd on the same branch. The Zenodo concept DOI for all versions is 10.5281/zenodo.22996790, and version 1.6.0 is 10.5281/zenodo.23082910. Version 1.7.0, which adds PREREG36, PREREG37 and the 1 October

notes, is 10.5281/zenodo.23085773. Version 1.8.0, which adds PREREG38 to PREREG44 and the second 5 October notes, is 10.5281/zenodo.23168366. Version 1.9.0, which adds PREREG45, PREREG45b, the stress tests and the fourth 5 October revision, was published on October 26, 2026 as 10.5281/zenodo.23665173 (its record metadata carries the staging date, October 5). Version 1.10.0, which adds the cross-version benchmark, PREREG46, the PREREG47 reading, the 14.2 arms, PREREG48, PREREG49 and this revision g, was published on October 27, 2026 as 10.5281/zenodo.23670727. The benchmark ran on OSC Cardinal as the 13.1h hindcast array 15578311, the 200 yr array 15578378 and the post-harvest array 15578563, and it was scored by analysis job 15579011 and its label only revision 15579160, with every launcher, runner copy and runner diff in `docs/benchmarks/2026-10-16_cross-version/scripts/`. The per stand frames, class file and results name plots and stands, so they stay on Cardinal under `~/acdx_20261016`. PREREG46 (md5 eb4b753d) and PREREG47 revisions 1 to 5 (7f64d22c, 6a77fa5f, 21a36a30, 4fa7aea1 and d93f3035) were registered before any of their runs; `thresholds_p47.R` wrote its thresholds (md5 e9609cfb) before the reading, and the PREREG46 assembly md5 (5d593fca) was recorded after a dry run and before the assembly. PREREG48 revision 1 (61ab907e) was registered before its scripts; its RESULT was revised once with a verification addendum (06807ec6, earlier text kept). PREREG49 revision 1 (c2188088) was registered after five server lookups and before any PREREG49 script.

1.7.0 DOI: 10.5281/zenodo.23085773

1.8.0 DOI: 10.5281/zenodo.23168366

1.9.0 DOI: 10.5281/zenodo.23665173

1.10.0 DOI: 10.5281/zenodo.23670727

Summary of studies

Track	Question	Verdict	Key number	RESULT file
PREREG16	Volume scored with the same height convention on both sides	3% rule fires, 14.2 volume gates use C1 or C2	C0/C1 gap over 3% in 12 of 20 sets	2026-09-27_v142_volume-scoring-convention_PREREG16_RESULT.md
PREREG17	NS IMULT under threshold matching, scored under C1	Not adopted	m* = 2.5, E3 and E6b fail	2026-09-27_v142_ns-ingrowth-multiplier-C1_RESULT.md
PREREG18	NS IMULT refined on a held out C1 volume criterion	Not adopted (edge)	m* = 1.5, fold B ratio 1.0074	2026-09-28_v142_ns-imult-refine_RESULT.md
PREREG19	QC anchor QC_AFAC 0.030	Not adopted (flat), C2	fold A C1 span 0.0067	2026-09-28_v142_qc-

Track	Question	Verdict	Key number	RESULT file
	to 1.000 under C1	withdrawn		anchor-C1_RESULT_r3.md (revision 3)
PREREG20	Two segment NS height form, 25 cm and over	Not adopted alone, E6b fails	ns_c fold B ratio 1.250 to 0.792	2026-09-28_v142_ns-height-form-25cm_RESULT.md
PREREG21	Why MNAP State model heights run tall	Diagnostic, no engine change	32% of t1 trees lack a measured height	2026-09-28_v142_mnap-state-height_RESULT.md
PREREG22	PREREG20 form with IMULT 1.5	Every gate passes, 14.2 NS candidate	interaction terms below 0.002	2026-09-29_v142_ns-height-by-ingrowth_RESULT.md
PREREG23	QC anchor refit on tag height increment, 1.061	Not adopted; under field tags every gate passes (PREREG42 erratum), not reopened	25 cm and over worse by 0.077 m yr^{-1} (row index), better by 0.108 (field tags)	2026-09-29_v142_qc-anchor-hinc_RESULT.md
PREREG24	QC stems and QMD diagnostic	Tables only, no verdict	none cited	not found on the host
PREREG25	NS small tree height increment below 12.7 cm	Program effect, no size multiplier	NS_RS under predicted by about 40%	2026-09-29_v142_ns-small-tree-height_RESULT.md
PREREG26	Bakuzis NS grid, v141LTfi against 14.1	Not cleared	BA ordering margin 8 against 7	2026-09-30_v142_bakuzis-ns-v141LTfi_RESULT.md
PREREG27	NS IMULT by remeasurement interval	Survivor loss, no interval term	ns_c survivor ratio 0.868	2026-09-30_v142_ns-interval-imult_RESULT.md (revision 2)
PREREG28	QC anchor by	Every gate	TOPHT +11.64%	2026-09-

Track	Question	Verdict	Key number	RESULT file
	size, 1.265 and 0.030	passes, fit on row index pairs (PREREG37)		30_v142_qc-anchor-size_RESULT.md
PREREG28b	QC anchor 1.265 below 25 cm, 0.111 kept	Every gate passes; not treated as candidate in these notes (fit on row index pairs, PREREG37)	TOPHT difference 0.33 points	2026-09-30_v142_qc-anchor-small-only_RESULT.md
PREREG29	NS_RS stand state, PREREG20 form refit on NS_OP	Refit fails A1 and A2, program effect	best NS_RS quartile 0.82	2026-09-30_v142_nsrs-state-and-nsop-refit_RESULT.md
PREREG30	NS form on long remeasurement intervals	Every gate passes	ns_long_c level 1.0152 to 1.0041	2026-09-30_v142_ns-long-interval-height_RESULT.md
PREREG31	Bakuzis QC grid for p28b_small	Not cleared	Reineke margin 1.496 against 1.266	2026-09-30_v142_bakuzis-qc-p28b_RESULT.md
PREREG32	Stands with an empty observed t2 list	No defect, admission kept, checks proposed	33 of 355 nb_c stands	2026-09-30_v142_nb-empty-t2-audit_RESULT.md
PREREG33	QC anchor reassigned from current DBH each year	Benchmark passes, Bakuzis not cleared, built on row index pairs (PREREG37)	Reineke margin 1.412 against 1.266	2026-10-01_v142_qc-anchor-dynamic_RESULT.md
PREREG34	ns_long_fl 10 to 19 year height deficit	Program effect	bin is 62% NS_RS trees	2026-10-01_v142_ns-long-fl-mid-interval_RESULT.md
PREREG35	Scorer r2 with	S0 fails on QC C2	48 lines added, 0	2026-10-

Track	Question	Verdict	Key number	RESULT file
	defect, exclusion and stratum checks	only, no adoption verdict	removed	01_v142_scorer-checks-r2_RESULT.md
PREREG36	QC C2 tag identity and adoption of r2	Tag defect found, r2 adopted from PREREG37, height tag key not covered by its search (PREREG37)	38.6% of qc_c row index pairs are the same tree	2026-10-01_v142_c2-tag-identity-r2-adoption_RESULT.md
PREREG37	What in the 14.1 QC height increment produces the size dependent anchor	Row index artifact; field tag refit is a new track (done, PREREG38)	field tag R at 25 cm and over 0.943, single QC_AFAC near 1.17	2026-10-01_v142_qc-height-structure_RESULT.md
PREREG38	Single QC anchor refit on field tags	Not adopted, TOPHT gate fails	QC_AFAC 1.242, pooled ratio 1.053, TOPHT +16.21% against +14.63%	2026-10-01_v142_qc-anchor-fieldtag-refit_RESULT.md
PREREG39	Decomposition of the TOPHT gate against the dominant cohort	Observed height rule dominates	9.29 of 16.78 points, 60.6% of heights imputed	2026-10-02_v142_topht-gate-decomposition_RESULT.md
PREREG40	TOPHT gate on measured heights	1.242 within the 10% ceiling (FIA length MQO, confirmed October 5)	+1.60% (1.242) against -8.45% (0.111)	2026-10-02_v142_topht-measured-gate_RESULT.md
PREREG41	Bakuzis QC grid for 1.242 and a matched tree TOPHT bound	Not cleared; measured top height reading robust	Reineke margin 1.490 against 1.266; matched tree +1.50%	2026-10-02_v142_bakuzis-qc-p38fit_RESULT.md
PREREG42	Erratum for the row index height gates, PREREG19 revision 3, QC close out	PREREG23 Rule 3 to Rule 6, others unchanged; 0.111 kept	11,632 (1,569) row index against 10,226 (2,299) field tags	2026-10-02_v142_rowindex-erratum_RESULT.md

Track	Question	Verdict	Key number	RESULT file
PREREG43	Where the shipped QC anchor 0.111 came from (records only, with addenda 2 and 3)	No fit of 0.111 itself; set by hand from the rounded printed means of a stand level increment diagnostic	ratio 8.888861; 0.111 = 0.018 / 0.162 as printed	2026-10-05_v142_qc-anchor-0111-provenance_RESULT.md, _RESULT_addendum2.md and _RESULT_addendum3.md
PREREG44	Bakuzis QC grid from measured stand lists, 1.242 against 0.111, informs 14.3 only	Not cleared	Reineke margin 1.101 against 0.768; top height at 100 years 28.01 against 17.29 m	2026-10-05_v142_bakuzis-qc-standlists_RESULT.md
PREREG45	Relative height multiplier exp(a + b HT/ESI) for the QC height increment, informs 14.3 only	Every benchmark gate passes; not cleared on long run grounds (frontier bound, Reineke)	fold B ratio 1.003; median top height over ESI 1.309 at 100 years against 1.175	2026-10-05_v142_qc-relht-increment_RESULT.md (revision 2, with stress tests ST1 to ST4)
PREREG45b	NB relative height multiplier at stand level (no stable NB tags), informs 14.3 only	Not adopted (Rule 2); NB measured top height too noisy a target	fold B ratio 0.556 against 0.90 to 1.10; 14.1 0.281	2026-10-05_v142_nb-relht-increment-standlevel_RESULT.md
Cross-version benchmark	13.1h, 14.0.4 L2 and 14.1 on 19 sets, post-harvest Maine FIA and 200 yr projections (not preregistered)	Diagnostic, no engine change; p38_fit and qc_rh run October 26, identical to 14.1	14.1 heavy removal recovery +1.39 Mg ha^{-1} yr^{-1}; median 19 set AGB bias 3.8% (14.1) against 9.6% (13.1h)	docs/benchmarks/2026-10-16_cross-version/README.md
PREREG46	QC height increment multiplier with a zero in relative height (qc_lz, qc_ez), informs 14.3 only	No candidate: qc_ez fit stop (Rule 1); qc_lz not adopted (Rule 2), gate C and Reineke also fail	qc_lz zero at r 1.635; 1 to 1.25 bin ratio 1.341 (upper 2.680)	2026-10-26_v142_qc-relht-zero-form_RESULT.md

Track	Question	Verdict	Key number	RESULT file
PREREG47	Absolute Bakuzis gates on the measured QC stand lists, reading r1, informs 14.3 only	Every arm read not credible on the long run (Rule 2)	qc141 late growth 7 of 36; offset removed median 1.418, 1.205 and 1.182 for p38_fit, qc_rh and qc_lz against 1.175	2026-10-26_v142_qc-bakuzis-absolute_RESULT.md
PREREG48	QC stems deficit split into survivors, upgrowth, recruitment and residual on field tags, informs 14.3 only	Rule 3: 14.3 ingrowth track on 14.1 and qc_rh, AR passes; verified independently	ingrowth share 1.269 and 1.251; survivor ratio 1.094 and 1.089; engine ingrowth below the floor	2026-10-26_v142_qc-stems-deficit_RESULT.md
PREREG49	NS gamma frailty survival step, and v141LTfi on the NS grid from measured stand lists, informs 14.3 only	Stage 1 rule 2 on M3, not adopted (M4 and M5 also fail); stage 2 v141LTfi not cleared (Eichhorn, top height monotonicity)	survivor ratio 0.930 and 0.993; 100 yr top height loss median 15.6%	2026-10-27_v142_ns-survival-step_RESULT.md